Impacts of Ocean Alkalinity Enhancement on the Seasonal Cycle of CO_2 ~~Flux~~ and Ocean pCO_2 in European Waters

Chiara Ciscato, David P. Keller

Key Points:

- Ocean alkalinity enhancement dampens the ocean pCO_2 seasonal cycle
- Ocean alkalinity enhancement amplifies the CO_2 flux seasonal cycle
- Regions like shallow, well-mixed waters of the southern North Sea ~~are~~ ideal to implement ocean alkalinity enhancement

Corresponding author: Chiara Ciscato, `cciscato@geomar.de`

Abstract

To meet the Paris Agreement, emission reduction must be complemented by carbon dioxide removal technologies aiming to sequester atmospheric CO_2 and store it in permanent natural reservoirs. One ~~of these~~ method~~s~~ is ocean alkalinity enhancement (OAE), which works to lower ocean pCO_2 and accelerate ~~the natural~~ CO_2 sequestration ~~through chemical weathering~~. As seasonality is a fundamental component of the ocean net annual CO_2 uptake, this study ~~contributes to bridge~~ the knowledge gap on the impacts of OAE on the seasonal carbon cycle. By analysing Earth System Model simulations of OAE performed at the European coastline, it was found that: with alkalinity addition, the ocean pCO_2 seasonal cycle is dampened due to its decreased sensitivity ~~to~~ CO_2 fluctuations; the CO_2 seasonal flux is amplified due to the larger imbalance at the air-sea interface in an alkalinised ocean; while the former outcome is stronger under low climate warming, the latter is stronger under higher warming.

Plain Language Summary

As global warming is largely caused by increasing atmospheric CO_2 concentration from human activities, ~~tech~~nologies are being developed to sequester excess carbon dioxide from the air. Among the proposed CO_2 removal options is ocean alkalinity enhancement (OAE). This method involves the addition of alkaline minerals or solutions at the ocean surface to lower the CO_2 concentration in seawater and accelerate its uptake from the atmosphere. As seasonality is important to understand the annual carbon movement, this research helps define the impacts of OAE on the seasonal CO_2 cycle. A numerical earth system model simulated continuous alkalinity addition at the European coastline from 2015 to 2100, where it was found that: adding alkalinity decreases the ocean pCO_2 seasonal amplitude by reducing its sensitivity to system changes, and ~~that~~ it increases the seasonal amplitude of CO_2 flux as the air-to-ocean CO_2 concentration gap becomes larger.

1 Introduction

1.1 Research context

Owing to its physical, chemical, and biological properties, the ocean constitutes a valuable resource for climate stabilisation, absorbing much of the earth's excess heat and storing vast amounts of carbon in its interior and deep waters (Friedlingstein et al., 2022; Scott-Buechler & Greene, 2019). With such premises, research is expanding on marine Carbon Dioxide Removal (CDR); that is, the employment of wetlands, bank shelves and the open ocean for carbon dioxide sequestration and storage. One of the marine CDR under investigation is ocean alkalinity enhancement (OAE), which aims to speed up the natural ocean CO_2 uptake through accelerated chemical weathering.

Alkalinity is an ocean property that describes the excess of bases, or proton acceptors, over acids, or proton donors (Wolf-Gladrow et al., 2007). Theoretically, OAE would be performed by releasing alkaline minerals or solutions at the ocean surface. Chemically, this operation would convert aqueous CO_2 (CO_2(aq)) into stable bicarbonate (HCO_3^-) and carbonate (CO_3^{2-}) ions, thus lowering ocean CO_2 partial pressure (pCO_2) and allowing for additional CO_2 uptake from the atmosphere to restore the altered equilibrium. While increasing the ocean buffering capacity, OAE would have the additional benefit of counteracting intensifying ocean acidification through the bonding with free hydrogen ions (H^+) in seawater.

Although OAE has been widely investigated to account for theoretical benefits and challenges, little attention has been given to its effects on the seasonality of the carbon cycle, which is a major component of the annual net ocean uptake. OAE implementation is likely to trigger significant seasonal impacts. First, if changes to the CO_2 flux in one direction are not counterbalanced by compensating effects in the opposite direction, these asymmetries may encourage modifications to the annual net cycle (Fassbender et al., 2022). Furthermore, as CO_2 flux seasonality is a central driver of ocean extreme events such as earlier calcium carbonate undersaturation (Kwiatkowski et al., 2022) and intensified ocean acidification (Sasse et al., 2015), other than acting upon inter-annual variability patterns (Rustogi et al., 2023), unknown looping mechanisms may affect the marine ecosystem at different scales. Lastly, while atmospheric pCO_2 responds to system stimuli at much shorter timescales, the ocean can take several seasons to react (Salt et al., 2013) and ~~defining~~ such time ~~lapses~~ is necessary for future OAE application.

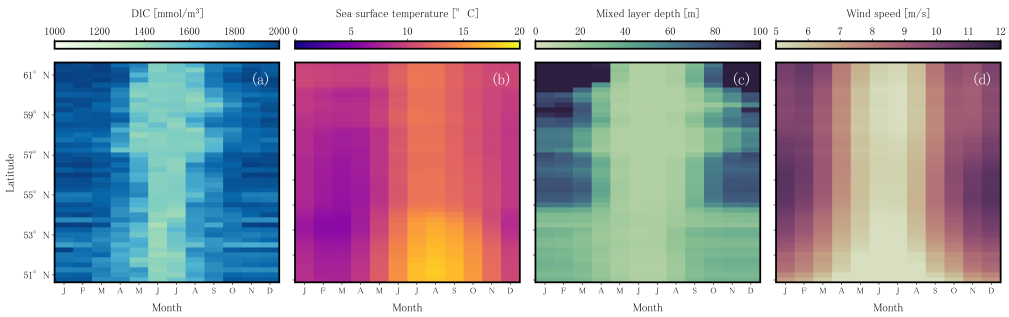
All these unexplored mechanisms leave a research vacuum that this study wishes to fill through the analysis of output data from an Earth System Model (ESM) that ~~performed~~ OAE along the European coastline in both a low-emission (SSP1-2.6) and a high-emission

(SSP3-7.0) pathway. ~~By doing so, we wish to~~ (a) define the impacts of OAE on the seasonal cycle of CO_2 flux and ocean pCO_2; (b) evaluate the influence of the background climate scenario on such outcomes; (c) quantify OAE efficacy in European waters.

1.2 Mechanisms driving CO_2 seasonality in the study area

Seasonality is a fundamental property of the annual net carbon cycle (Rodgers, 2023; Fassbender et al., 2022). Although with regional and local variations, ocean CO_2 seasonal flux is mainly controlled by two factors: sea surface temperature (SST) and Dissolved Inorganic Carbon (DIC). As SST acts on gas dissolution in seawater, in winter months, colder temperatures favour CO_2 solubility while warming water in summer leads to CO_2 outgassing (Williams & Follows, 2011). As for DIC-driven seasonality, in winter, where processes like upwelling and organic matter (OM) respiration prevail and supersaturate the DIC pool, the ocean easily loses CO_2 to the atmosphere, whereas summertime OM production and downwelling favour atmospheric CO_2 drawdown. As the dynamics induced by SST and DICs oppose each other (Lerner et al., 2021; Takahashi et al., 2002), the one prevailing drives the direction of seasonal CO_2 exchange at a given location (Sarmiento & Gruber, 2006).

Figure 1: From left to right, latitudinal transect of the North Sea (51°N-62°N) for the baseline seasonal cycle of dissolved inorganic carbon (DIC) (a), sea surface temperature (SST) (b), mixed layer depth (MLD) (c), and wind speed (d) averaged over 2015-2100 in SSP1-2.6.



The air-sea CO_2 flux in European waters, for example, is characterised by a strong North-to-South seasonal gradient (Salt et al., 2013; Prowe et al., 2009). Figure 1 offers a representation of some of the features that affect CO_2 flux in the North Sea. Biological drivers dominate in the well stratified northern North Sea (fig. 1a), where a relatively shallow mixed layer allows for nutrient-rich waters to stay at the surface and encourage phytoplankton bloom during the productive season. OM that there accumulates then sinks

below the surface layer, where remineralisation can happen far from atmospheric contact. At this location, CO_2 sequestration through carbon fixation outweighs temperature-driven CO_2 outgassing, making the area a strong carbon sink (Salt et al., 2013; Prowe et al., 2009).

Conversely, in the southern North Sea and the adjacent European continental shelf, where a shallow water column, wind and tidal forces (fig. 1d) generate a more uniform layer all year round (fig. 1c) (Thomas et al., 2009; Hoppema, 1990), the air-sea CO_2 exchange is mainly driven by temperature variability (fig. 1b) (Salt et al., 2013). Carbon production and remineralisation happen within the same surface compartment, where CO_2 can easily be exchanged with the atmosphere (Prowe et al., 2009; Thomas et al., 2004). At this location, community respiration outweighs photosynthetic activity (Artioli et al., 2012), resulting in a region of weak CO_2 outgassing (Prowe et al., 2009).

2 Methods

2.1 Earth System Model

The data used in this manuscript was derived from the Flexible Ocean and Climate Infrastructure (FOCI), an emission-driven ESM of which a detailed description is provided by Matthes et al. (2020). The model consists of an atmosphere, an ocean and a land surface component. The atmosphere is divided into 95 vertical layers and has a horizontal grid resolution of ∼1.8°. The ocean is composed of 46 vertical layers and has a horizontal resolution of 1/2°.

FOCI incorporates the Model of Oceanic Pelagic Stoichiometry (MOPS) to enable the simulation of marine biogeochemical cycles, including a full carbon cycle. This allows to resolve biological dynamics such as phytoplankton uptake and remineralisation, calcite formation and dissolution. FOCI-MOPS underwent a 500-year spin-up, after which steady state was achieved for all variables, with the commonly-observed exception of nitrogen (Chien et al., 2022).

Ocean pCO_2, that in FOCI is termed surface CO_2 fugacity (fCO_2), is calculated as follows:

$$fCO_2 = CO_2 \cdot / K_0 \tag{1}$$

where $CO_2 \cdot$ is the surface ocean dissolved CO_2 concentration in mol kg^{-1} and K_0 is the seawater solubility coefficient of CO_2 in mol kg^{-1} $\cdot$ atm as prescribed by Weiss (1974). Both factors are affected by a multitude of intertwined physical and biological processes which are sometimes difficult to analyse individually. Some of the variables may also be subject to over and underestimation, therefore altering connected interactions (Chien et al., 2022).

The parameterisation of air-sea gas exchange in FOCI is described by the following equation:

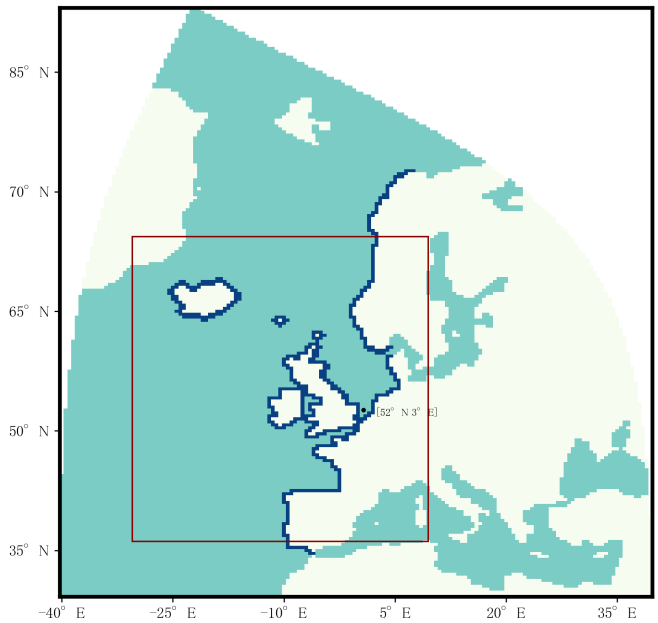
$$F_{CO_2} = k_w([CO_2 \cdot]_{sat} - [CO_2 \cdot]) \tag{2}$$

where k_w is the gas transfer velocity in m s^{-1}, and $CO_2 \cdot (sat)$ is the saturation concentration of CO_2 in mol kg^{-1} (Chien et al., 2022). It is important to highlight that CO_2 flux calculations are performed in the atmospheric component of FOCI, revealing a coarser geographical resolution.

2.2 Experimental design

Historical simulations were run from year 1850 to year 2015, from which the scenarios begin, establishing the baselines for the analysis. The first warming pathway is SSP1-2.6 and describes a low-emission future with a radiative forcing of 2.6 W m^{-2} by the end of the century. The second is SSP3-7.0 and narrates a high-emission scenario with a radiative forcing of 7.0 W m^{-2} by 2100.

Figure 2: Geographical OAE application. Blue bordering line represents alkalinity addition and red rectangle defines the cropped area used in this paper. Black circle is the location of point S



In FOCI, alkalinity is a prognostic tracer simulated as a combination of biological sources and sinks: nitrate and phosphate supply as sinks, calcium carbonate production and dissolution as sink and source, respectively, and OM formation and remineralisation, as sink and source, respectively (Chen et al., 2021).

OAE is performed using a masking approach from year 2015 to year 2100. Alkalinity addition is subject to a linear increase over the first simulation decade (2025-2035), until the equivalent of 1 Gt yr^{-1} of fast-reacting calcium hydroxide ($Ca(OH)_2$) is reached.

$$Ca(OH)_2 + 2CO_2 \leftrightarrow Ca^{2+} + 2HCO_3^- \tag{3}$$

$Ca(OH)_2$ consumes two moles of atmospheric CO_2 and produces one mole of calcium (Ca^{2+}) and two moles of bicarbonate, therefore ideally increasing alkalinity by a factor of two (eq. (3)) (Chien et al., 2022). Alkalinity is added to the system continuously and uniformly along the European coastline of 50-kilometre width, with the exclusion of the Baltic and the Mediterranean seas which are poorly resolved by the model (fig. 2).

2.3 Data processing

The global output data delivered by FOCI was sliced to the European region, with coordinates -25° to 10° E and 40° to 70° N (red rectangle in fig. 2). Although the cropped area does not cover the entire European region where OAE was performed, and part of the Norwegian and Spanish littoral were left out of the analysis, this approximation does not result in a meaningful modification from the modelled regional definition.

A comparison between the European region and location S set at 52°N and 3°E (black circle in fig. 2) is able, on the one hand, to determine an approximated estimate of how the European system as a whole would behave in the future and, on the other, to draw more precise conclusions on the regime shifts taking place at the epicentre of the domain.

Assuming that the most evident baseline-to-OAE disparities are detected at the end of the simulation, the annual cycle averaged over the last simulation decade (2090-2100) is provided. While ocean pCO_2 and CO_2 flux are estimated only for the sea surface, alkalinity is measured over all 46 FOCI ocean layers. Therefore, alkalinity averages were vertically integrated over the mixed layer depth (MLD), which covers an ocean region of dynamic, uniform mixing, subject to geographical and seasonal variations.

3 Results

In this section, results are presented for alkalinity, ocean pCO_2, and CO_2 flux.

Figure 3: Annual cycle averaged over 2090-2100 for alkalinity (a, b, c, and d), for ocean pCO_2 (e, f, g, and h), and for CO_2 flux (i, j, k, and l), with all relative change (baseline - OAE). In CO_2 flux plots, negative values indicate ocean uptake.

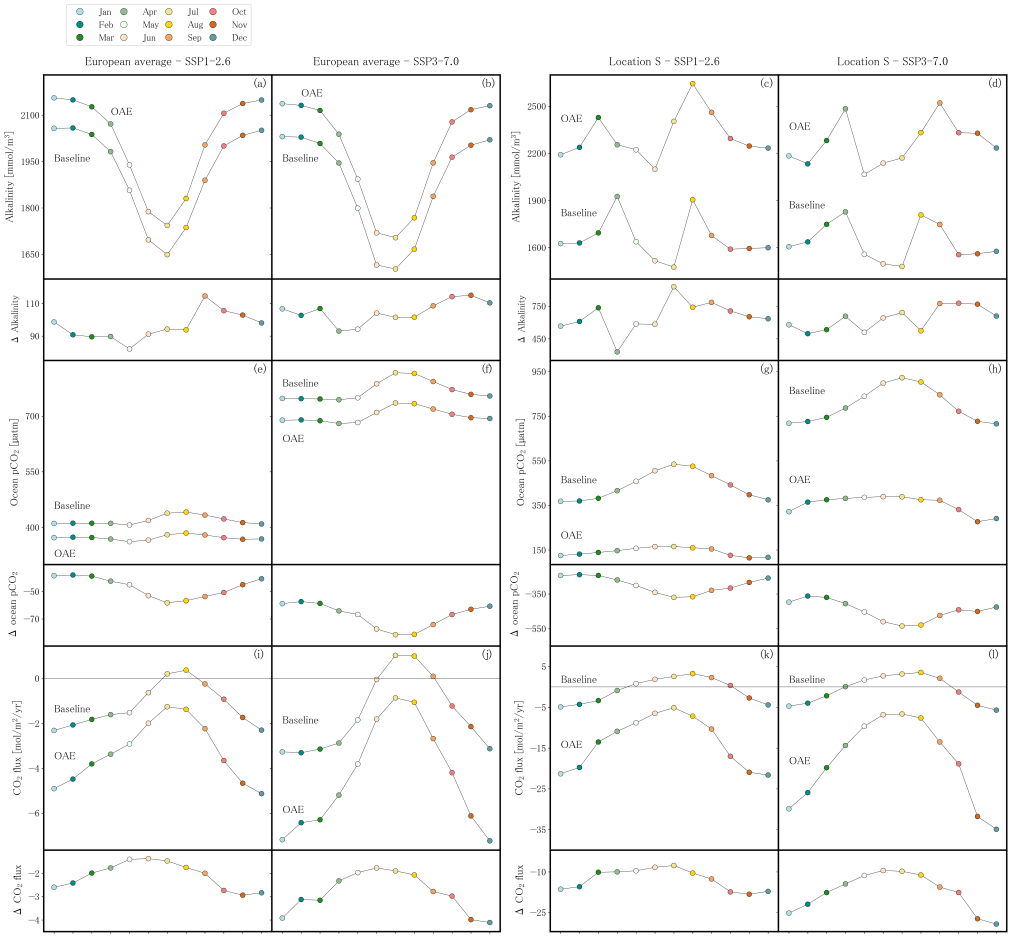


Figure 3 reproduces spatially-aggregated values over the European domain (first two columns) and at location S (last two columns) for both scenarios averaged over the last simulation decade, as well as the difference between the baseline and the OAE-induced scenario.

Figure 4: Seasonal amplitude change (baseline - OAE) averaged over 2090-2100 for alkalinity (a and d), for ocean pCO_2 (b and e) and for CO_2 flux (c and f). Black circle is location point S.

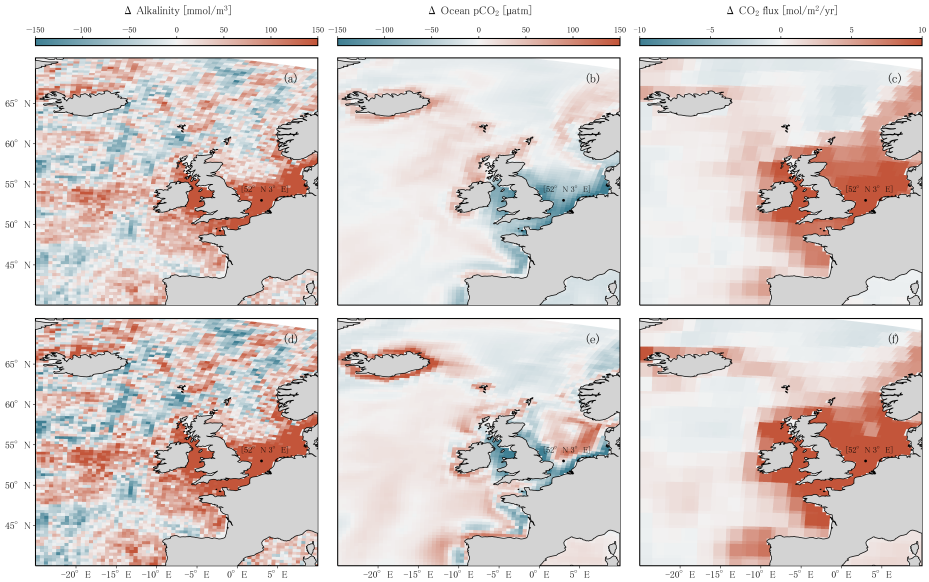


Figure 4 focuses on the spatial alterations induced by alkalinity addition by substracting the OAE-induced scenario from the baseline in SSP1-2.6 (first row) and in SSP3-7.0 (second row).

3.1 Alkalinity

~~At~~ the European average in SSP1-2.6 (fig. 3a), baseline alkalinity registers lowest values in summer and highest in winter, with respective extremes corresponding to 1649 and 2059 mmol m^{-3}. With alkalinity addition, absolute values rise at all seasons and the largest change between the two scenarios takes place in September. In SSP3-7.0 (fig. 3b), baseline alkalinity levels are slightly lower than the respective lower-emission pathway but, when OAE is implemented, absolute values rise to similar levels as SSP1-2.6.

At location S for the lower warming scenario, the baseline cycle is similar in both warming trajectories, with two high peaks in April and August, and lowest levels registered in July. Absolute values range between 1475 and 1927 mmol m^{-3} in SSP1-2.6 (fig. 3c), and between 1479 and 1828 mmol m^{-3} in SSP3-7.0 (fig. 3d). Under OAE-induced conditions, absolute values grow substantially and now range from 2099 (June) to 2645 (August) mmol m^{-3} under low warming, and from 2066 (May) to 2523 (September) mmol m^{-3} under high warming.

In both scenarios, the alkalinity spatial pattern reveals strongest variation in the southern North Sea, where the seasonal amplitude change increases greatly. Almost no difference is observed between SSP1-2.6 (fig. 4a) and SSP3-7.0 (fig. 4d) and no substantial modification is detected ~~at~~ open ocean.

3.2 Ocean pCO_2

~~At~~ the European average in SSP1-2.6 (fig. 3e), ocean pCO_2 is characterised by least and most elevated values in May and August, respectively, with a mean amplitude stretching from 407 to 441 µatm. In the OAE projection, ocean pCO_2 is reduced due to alkalinity enhancement and absolute values drop. The amplitude also decreases, now spanning from 362 to 385 µatm. Δ ocean pCO_2 ~~detects~~ the largest baseline-to-OAE anomaly in summer, meaning that alkalinity stimulates pCO_2 decline during CO_2 outgassing. In the baseline of SSP3-7.0 (fig. 3f), pCO_2 maintains an ascending trend until 2100 (not shown). Elevated values are recorded in July, to then decline until minima are reached in April. The mean seasonality has lowest and highest peaks of 744 and 817 µatm, respectively. OAE induces the seasonal cycle to decrease substantially, with an average amplitude of 55 µatm.

Unlike the European average, location S displays minima in winter under both warming scenarios, denoting that pCO_2 decrease is here anticipated. OAE reduces pCO_2 absolute values, with a mean drop of 200 µatm in SSP1-2.6 (fig. 3g) and of 400 µatm in SSP3-7.0 (fig. 3h). OAE contributes to ocean pCO_2 decrease especially over the warm months, with a significant seasonal amplitude reduction: in the low warming scenario, it is compressed by more than a third compared to the baseline, shifting from 166 to 51 µatm, while in SSP3-7.0 it is almost halved, moving from 207 to 111 µatm.

In SSP1-2.6 (fig. 4b), a clear spatial pattern is identified at the proximity of the injection location, with two diverging geographies: at higher latitudes, in Iceland and in the Norwegian and British North-West bank, pCO_2 seasonal amplitude is slightly more pronounced than the baseline. Conversely, at lower latitudes including the mainland shoreline and the central-to-southern UK coasts, the pCO_2 seasonal cycle is strongly mitigated.

In SSP3-7.0 (fig. 4e), pCO_2 amplitude behaves similarly and maintains the spatial pattern outlined above, with the exception of the central North Sea, where strong amplification is displayed. ~~At~~ open sea, pCO_2 seasonality is slightly amplified in the North Atlantic and dampened in the sub-polar regions. Iceland represents a deviation from the expected outcome, as it displays a strong seasonal amplification signal with OAE application.

3.3 CO_2 flux

Figure 3i shows that, ~~at~~ the European average in SSP1-2.6, the baseline ocean tends to outgass CO_2 in summer (July and August), while remaining a carbon sink over the remaining months, with most pronounced uptake in winter, especially December. When OAE is applied, the system is turned into a sink all year round, with summertime (wintertime) still reflecting the weakest (strongest) CO_2 uptake. The seasonal amplitude varies noticeably, with the OAE scenario showing a much greater breadth than the baseline. OAE-induced CO_2 flux amplitude has an average span of -1.25 to -5.12, compared to the baseline that ranged between 0.37 and -2.31 mol m^{-2} yr^{-1}. In SSP3-7.0 (fig. 3j), the baseline outgasses CO_2 from July to September. Adding alkalinity increases the CO_2 seasonal amplitude more than the low emission trajectory, especially over the cold months.

In the baseline of both scenarios at location S (fig. 3k, l), the annual cycle is split into two equally long phases. Outgassing happens from May to September (SSP1-2.6) and October (SSP3-7.0) and uptake from November to March (SSP1-2.6) and April (SSP3-7.0). In OAE-driven SSP1-2.6, CO_2 flux fluctuates between -21.6 and -5.11 mol m^{-2} yr^{-1} on average, therefore doubling the baseline amplitude, while seasonality of CO_2 flux in SSP3-7.0 is three times larger than the baseline, with a current amplitude of 28 mol m^{-2} yr^{-1}. In all Δ CO_2 graphs, the baseline-to-OAE discrepancies grow in winter and decrease in summer, meaning that the impacts of alkalinisation on the ocean surface are largest when ocean uptake is most pronounced.

With OAE, SSP1-2.6 (fig. 4c) reveals a pronounced increase in the CO_2 seasonal flux all around the European continental coastline, and the signal remains strong until the connection with open ocean. ~~Poor~~-to-no changes are observed in Iceland and Northern Spain. SSP3-7.0 (fig. 4f) mirrors the same spatial pattern as SSP1-2.6. When OAE is implemented, although the North-West Icelandic coast reveals strong amplification.

4 Discussion

4.1 European average

All maps clearly show that system perturbations for ocean pCO_2 and CO_2 happen within the model domain, at the site proximity of alkalinity injection, and the signal does not propagate far by the end of the century. This leads to conclude that successful OAE deployment is spatially limited to where alkalinity is added, an outcome that is corroborated by a recent experiment carried out by Wang et al. (2023).

While open-ocean regions like Iceland, Norway and Spain show few-to-absent perturbations with alkalinity addition, the most affected area seems to be the North Sea, especially the South which delivers pronounced system alterations. This is likely due to the fact the shallower, non-stratified ocean system allows for alkalinity to accumulate at the surface and to not be lost due to mixing-induced subduction before equilibration is completed (Wang et al., 2023; Jones et al., 2014).

In both SSP1-2.6 and SSP3-7.0, ocean pCO_2 minima are registered in spring, which may be the result of phytoplankton bloom and consequent high photosynthetic activity, where CO_2 is consumed. In SSP3-7.0, substantial atmospheric CO_2 emissions force pCO_2 to rise progressively until the end of the simulation, even with alkalinity enhancement. With OAE application, ocean pCO_2 seasonal amplitude is reduced because an alkalinised ocean minimises pCO_2 sensitivity to DIC changes. However, regions like Iceland and Northern United Kingdom constitute an exception, as alkalinity addition causes an amplification signal of the ocean pCO_2 seasonal cycle.

In agreement with previous estimates (Schwinger, 2022), OAE works to amplify the CO_2 seasonal flux in both SSPs. This happens because OAE alters the air-to-sea equilibrium, which thus requires higher CO_2 input into the ocean to restore it. In both scenarios, highest levels of alkalinity are recorded between the end of summer and the onset of autumn but largest uptake takes place in winter, therefore introducing a short response lag due to re-equilibration timescales. In the OAE scenario, alkalinity addition prompts a greater CO_2 flux downward in wintertime, meaning that OAE induces an asymmetrical regime shift, with much lower lows than lower highs. This outcome may result in seasonally-dependent modifications to the local ocean's biogeochemistry, with unexpected looping mechanisms.

4.2 Location S

Air-sea re-equilibration can take from a few months to even years to complete, depending on the physical properties of the region (Jones et al., 2014). Considering the topographic characteristics of the southern North Sea, where a shallow water column combines

with strong wind-driven forces and tidal currents, a region of uniform mixing encourage fast re-equilibration with the atmosphere and allows for efficient carbon sequestration under enhanced alkalinity.

With OAE deployment, pCO_2 levels plummet. OAE-induced pCO_2 change is found to be highest in the season where the associated seasonal driver operates to reduce the ocean CO_2 sink potential. In this case, pCO_2 deviation from the baseline increases in summer, when rising temperatures mitigate gas solubility and inhibit ocean uptake. Under SSP3-7.0, OAE application is more successful than in SSP1-2.6 in lowering ocean pCO_2 absolute values, although seasonal compression remains stronger in the lower emission scenario.

At location S, CO_2 flux modulations are dominated by temperature changes (Rodgers, 2023; Salt et al., 2013; Prowe et al., 2009). Strongest CO_2 outgassing (and lowest uptake in the OAE-induced scenario) takes place in summer, when warm temperatures weaken solubility, whereas CO_2 uptake happens in winter. Stronger simulated global warming results in higher CO_2 ocean uptake as well as larger seasonal amplification caused by the enhanced air pCO_2 that forces the gas movement downwards into the ocean.

5 Conclusions

This study addressed the OAE-induced changes on the seasonal carbon cycle in European waters, therefore contributing to the understanding of CO_2 intra-annual variability when performing coastal alkalinity enhancement. It was found that: (a) OAE dampens the ocean pCO_2 seasonal cycle and it amplifies the CO_2 flux cycle; (b) the former outcome is stronger in SSP1-2.6 while the latter turns stronger in SSP3-7.0; (c) locations like the southern North Sea are ideal to implement OAE, as shallow waters of intense mixing fasten the CO_2 gas transfer at the air-sea interface.

Some limitations and recommendations for future research can be drawn from this analysis. Firstly, while freshwater input from rivers is modelled in FOCI, riverine alkalinity is not accounted for, limiting the reliability of resulting alkalinity absolute values. As river runoff features are deeply seasonal, implementing such component into future simulations should add complexity and precision to the system model, and thus enhance the understanding of external sources to the European seas.

Secondly, our analysis does not separate ocean pCO_2 into its thermal and biological element, therefore missing the influence that each of these have in altering the CO_2 seasonal cycle. Given the significance of temperature variability in the southern North Sea, differentiating between the two may provide further insight into the seasonal role of temperature and biology under OAE application.

Lastly, our OAE simulation, that is, continuous alkalinity addition along the whole European coastline, pictures a highly idealised scenario, which is unlikely to become a real-world case. Other, more realistic OAE techniques are being explored, such as alkalinity addition performed in pulses or at selected point locations (river mouths, sewage treatment plants, open ocean regions). The field of carbon cycle seasonal variability within OAE is still largely unexplored and further investigation is recommended to build the premises for conscious large-scale OAEapplication.

6 Open Research

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